

# Sector Economics and Go-to-Market Viability

When a F1R3FLY shard pays for itself, and which markets get it there

A companion to “Rent and Shard Splitting”

F1R3FLY Industries — working note

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*A companion note. It reuses the cost model, the dense-packing result, and the rent lifecycle of “Rent and Shard Splitting” [1] and asks the next question: per market sector, how much monthly traffic does a shard need to be economically viable, which sectors produce it, and in what order should we deploy. All dollar figures are engineering estimates on stated assumptions; every fee is a lever, flagged as such, so the tables re-derive when real schedules replace ours.*

## 1 The viability condition

A shard is viable when monthly phlogiston revenue covers its cost. From the rent note, one ten-node shard costs  $C \approx \$4,160/\text{month}$  all-in, and dense elastic packing [2] lets it carry on the order of 30,000,000 fee-bearing transactions per month before the split threshold binds. If a sector’s transactions bear a blended fee  $r_{\text{tx}}$  (precharge, activation, and rent combined, in dollars), then break-even monthly traffic is simply

$$T^* = \frac{C_{\text{eff}}}{r_{\text{tx}}}, \quad C_{\text{eff}} = C \times (\text{compliance multiplier}).$$

Risk pushes on *both* sides of this ratio. A riskier sector can charge a higher  $r_{\text{tx}}$ , but it also raises  $C_{\text{eff}}$ : replication, audit, and dedicated infrastructure break the “all co-resident on one cloud” assumption. The useful dimensionless quantity is therefore not  $T^*$  but the *break-even load*,  $T^*$  divided by per-shard capacity—the fraction of a shard a sector must fill to pay for it.

**Observation 1** (Capacity is never the gate; demand is.). *Every sector below is feasible on a single shard’s capacity—even the highest-volume case sits far under the split threshold per shard. Viability is gated entirely by whether a sector produces enough fee-bearing transactions, and the break-even load varies by more than two orders of magnitude across sectors. Riskier data is easier to make pay, not harder.*

**Common parameters.** Unless noted: base cost  $C = \$4,160/\text{shard}/\text{month}$ ; per-payload network cost at dense packing  $C/30,000,000 \approx \$0.00014$ ; low-risk shards run dense ( $\sim 28,000,000$ – $30,000,000$  tx/mo), mission-critical shards run with headroom ( $\sim 20,000,000$  tx/mo). Compliance multipliers: AI  $1.2\times$ , music  $1\times$ , biotech  $1.5\times$ , fintech  $2.5\times$ , medical  $4\times$ .

## 2 AI (ASI Alliance) — the growth engine

Assumptions:  $r_{\text{tx}} = \$0.01$  (inference calls, tool invocations, and agent-to-agent settlement all bear real value);  $C_{\text{eff}} = \$4,992$ ;  $\sim 10,000$  tx per agent per month at machine pace; dense packing.

**Break-even:**  $\sim 500,000$  tx/month =  $\sim 1.8\%$  of a shard. An AI shard pays for itself at two percent utilisation.

**Table AI.** Agent population ramp at  $10\times$ /year (unbounded).

| Year | Agents    | Monthly tx | Shards | Cost      | Revenue   | Margin      |
|------|-----------|------------|--------|-----------|-----------|-------------|
| 1    | 100       | 1.0M       | 1      | \$4,992   | \$10,000  | $2.0\times$ |
| 2    | 1,000     | 10M        | 1      | \$4,992   | \$100,000 | $20\times$  |
| 3    | 10,000    | 100M       | 4      | \$19,968  | \$1.0M    | $50\times$  |
| 4    | 100,000   | 1.0B       | 36     | \$179,712 | \$10M     | $56\times$  |
| 5    | 1,000,000 | 10B        | 358    | \$1.79M   | \$100M    | $56\times$  |

The defining trait is the absence of a TAM ceiling. Machine-generated volume is unbounded and sticky; margin saturates near  $56\times$  and then revenue simply scales. AI both load-tests the dispatch/COMM hot path under real volume *and* is the only sector whose growth does not terminate—which is why it should anchor the rollout rather than a low-fee consumer sector.

### 3 Fintech (Boring Financial) — the density engine

Assumptions:  $r_{tx} = \$0.25$  blended (settlement, contract-lifecycle, and loan-servicing events);  $C_{eff} = \$10,400$  (custody, audit, regulatory load); safety headroom.

**Break-even:**  $\sim 42,000$  tx/month =  $\sim 0.2\%$  of a shard—the most revenue-dense sector; a fintech shard pays for itself at one-fifth of one percent utilisation.

**Table FIN.** Transaction ramp.

| Year | Monthly tx | Shards | Cost      | Revenue   | Margin      |
|------|------------|--------|-----------|-----------|-------------|
| 1    | 50k        | 1      | \$10,400  | \$12,500  | $1.2\times$ |
| 2    | 500k       | 1      | \$10,400  | \$125,000 | $12\times$  |
| 3    | 5M         | 1      | \$10,400  | \$1.25M   | $120\times$ |
| 4    | 50M        | 3      | \$31,200  | \$12.5M   | $400\times$ |
| 5    | 500M       | 25     | \$260,000 | \$125M    | $481\times$ |

**Remark 1** (The most assumption-sensitive number in the note.).  $\$0.25/tx$  is right for high-value settlement events, where it is trivial against notional, and wrong by one to two orders of magnitude for high-frequency micropayments, where fintech behaves like AI on margins. The real schedule is basis-points-on-notional for large events and sub-cent for small ones, so fintech is bimodal; model the two instrument classes separately once Boring supplies their mix.

### 4 Medical (VA) — the cohort-capped annuity

The Veterans Health Administration serves roughly 9,100,000 enrolled veterans [5]; the 80% target is  $\sim 7,280,000$  patients. Assumptions:  $\tau = 50$  provenance-logged record accesses per patient per month;  $R = \$1/\text{patient/month}$  in platform fees (a conservative 2–7% of full-EHR per-patient spend);  $C_{eff} = \$16,640$ ; headroom.

**Break-even:**  $\sim 16,640$  patients against a  $\sim 400,000$ -patient shard =  $\sim 4\%$  load. A medical shard pays for itself at four percent utilisation.

**Table MED.** 10×/year rollout against a finite cohort (saturates at year 4).

| Year | Patients  | Monthly tx | Shards | Cost      | Revenue     | Margin       |
|------|-----------|------------|--------|-----------|-------------|--------------|
| 1    | 7,280     | 0.36M      | 1      | \$16,640  | \$7,280     | 0.4× (loss)  |
| 2    | 72,800    | 3.6M       | 1      | \$16,640  | \$72,800    | 4.4×         |
| 3    | 728,000   | 36M        | 2      | \$33,280  | \$728,000   | 21.9×        |
| 4    | 7,280,000 | 364M       | 19     | \$316,160 | \$7,280,000 | 23.0×        |
| 5–10 | 7,280,000 | 364M       | 19     | \$316,160 | \$7,280,000 | 23.0× (flat) |

Two facts dominate. First, the cohort is a hard wall: a literal 10×/year ramp saturates the finite population by year 4, so medical revenue is *logistic*—explosive then flat—and past saturation it is an annuity (~\$87M/year revenue on ~\$3.8M/year infrastructure), not a growth story. Cumulative decade revenue is ~\$620M, of which infrastructure is a ~4% rounding error. Second, the constraint is never economic: even tripling cost for high-availability replication (3××4× loading) leaves ~7–8× margin at cohort. The binding constraints are regulatory and operational.

**Remark 2** (Risk sets the rent posture.). “COMMs *delivered while funded*” (*the rent note’s mortality semantics*) is a safe default for music but a safety failure for a patient-record access that silently never fires on fund depletion. Mission-critical residuals must be pinned to committed on-chain balance (*germ line*), never allowed to deplete to eviction. This is the strongest reason medical deploys last: the rent model itself is most stressed exactly where the data is least forgiving.

## 5 Music — the cold-tail fill

A stable of ten independent artists at ~1M non-overlapping followers, each audient streaming a conservative ten songs/month, is 100M plays/month. Per-play on-chain, that is ~4 shards (\$16,640/mo), and break-even is

$$r_{\text{tx}}^* = \frac{\$16,640}{100\text{M}} \approx \$0.000166 \text{ per play.}$$

Any network fee above ~1/60 of a cent per play is profitable; at \$0.0003/play the stable runs ~1.8×.

**Table MUS.** Margin on the ten-artist stable (100M plays/mo, 4 shards, \$16,640/mo).

| Network fee/play | Monthly revenue | Cost     | Margin            |
|------------------|-----------------|----------|-------------------|
| \$0.0001         | \$10,000        | \$16,640 | 0.6× (underwater) |
| \$0.000166       | \$16,640        | \$16,640 | break-even        |
| \$0.0003         | \$30,000        | \$16,640 | 1.8×              |
| \$0.001          | \$100,000       | \$16,640 | 6.0×              |

Two design facts. Batching plays off-chain and settling (say 100:1) collapses 100M plays to 1M transactions on one shard at ~7× margin—strictly better economics *unless per-play provenance is the product*. If transparent per-play royalty and capability-attenuated rights are the differentiator over incumbents, we *want* the 100M on-chain transactions, and music becomes the abundant *cold-tail packing* that monetises the spare leaves of the AI and fintech shards (rent note, bid-rent stratification). Either way music is viable; the choice is provenance granularity versus margin. The

caveat is concentration: 100M/month resting on ten artists is roster-dependent and churns, which is why music is high-margin fill, never the anchor.

## 6 Web 2.0 — participatory advertising

Free-at-point-of-use consumer properties have no per-transaction user fee, so the framework above appears to fail—until advertising is made *participatory*. End users pay in tokenised impressions per data payload until their usage is paid off, then either switch to impression-free payloads or accumulate advertising tokens spendable in other markets. This both closes the monetisation gap and inverts the extraction relationship.

The mechanism rests on one ratio. A data payload costs \$0.00014 at dense packing; an ordinary display impression at a ~\$3 CPM [9] is worth \$0.003:

one ad impression covers the network cost of ~20 data payloads.

Because impressions are tokenised on-chain, each impression is itself a fee-bearing transaction that funds the payload it rides plus margin: the ad layer is self-funding volume, not overhead. When cumulative impressions exceed cumulative usage, the user crosses into the impression-free or net-earner fork.

**Table WEB.** Three Google properties; payloads as emails/sessions/edits. Volumes from Gmail [6], Maps [7], Workspace [8].

| Property                    | Payloads/mo | Shards   | Infra cost/mo | Break-even ads/user/day |
|-----------------------------|-------------|----------|---------------|-------------------------|
| Gmail (1,800,000,000 users) | ~3.6T       | ~120,000 | ~\$500M       | ~3                      |
| Maps (2,000,000,000 users)  | ~3.0T       | ~100,000 | ~\$416M       | ~2–3                    |
| Docs (3,000,000,000 users)  | ~600B       | ~20,000  | ~\$83M        | ~0.3                    |

Every property is funded if a user views two or three ads per day—far below real exposure—so the typical user crosses into accumulating spendable tokens. **That is the inversion:** in incumbent Web 2.0 the user is the product, their attention extracted and the surplus captured by the platform; here the user covers their own cost with attention and *keeps the surplus* as portable tokens. The network takes a thin per-payload margin at astronomical volume—a \$0.00002/payload take across Gmail’s 3.6T payloads is still ~\$72M/month. This is “the community is the moat” and the anti-enshittification thesis rendered as an income statement.

**Observation 2** (The cross-market spend saves Docs.). *Docs has the lowest break-even (~0.3 ads/day) but the highest ad resistance—nobody wants ads in documents. The fork resolves it: users earn tokens on high-CPM, ad-tolerant surfaces (Maps’ local and sponsored context) and spend them on impression-free Docs. The cross-subsidy is carried in the user’s portable tokens, structurally unlike incumbent bundling, where the cross-subsidy is the platform’s lever over the user.*

**Enter through a privacy wedge, not at Gmail scale.** 120,000 shards and \$500M/month is a hyperscaler buildout; per-payload viability is trivial but absolute capital is enormous. The wedge is a privacy-premium slice: 1% of Gmail (~18,000,000 users who actively want data ownership) is ~36B payloads/month ≈1,200 shards ≈\$5M/month infrastructure, trivially funded—and “own your data, earn from your attention” self-selects exactly those adopters. Web 2.0 is a beachhead-then-scale play entered once the participatory rails are proven on the earlier sectors.

## 7 Synthesis and rollout

**Table SUM.** The five sectors plus Web 2.0.

| Sector  | Revenue density          | Risk    | Boundedness       | Break-even load | Role                    |
|---------|--------------------------|---------|-------------------|-----------------|-------------------------|
| Web 2.0 | ad-funded (\$0.00014/pl) | low-mod | astronomical      | ~3 ads/user/day | mass-market moat (last) |
| Music   | \$0.0003/tx              | low     | unbounded, churny | ~near full      | cold-tail fill          |
| AI      | \$0.01/tx                | low-mod | unbounded, sticky | ~2%             | <b>anchor / growth</b>  |
| Fintech | ~\$0.25/tx               | high    | semi-bounded      | ~0.2%           | margin engine           |
| Medical | high/patient             | highest | hard cohort cap   | ~4%             | cash-cow annuity        |

The pattern that falls out: **revenue density and risk rise together, and both run opposite to volume.** The original safe-to-critical *risk* ordering is correct and unchanged. Layered on top, the *economic* ordering is:

1. **AI — anchor.** Simultaneous load-testing and viability; hardens the hot path; unbounded growth.
2. **Fintech — margin.** Highest revenue density rewards even modest volume; builds high-assurance discipline on financially recoverable failures.
3. **Medical — annuity.** Highest margin per unit, capped at the cohort, deployed last so the assurance and germ-line-pinning machinery is battle-tested first.
4. **Music — continuous fill.** Not a phase but the cold-tail packing that monetises spare capacity across the above.
5. **Web 2.0 — mass-market substrate.** Entered through a privacy wedge once the participatory ad rails are proven.

## 8 Sensitivity and open levers

Every figure scales transparently. The fees  $r_{tx}$  are the dominant levers and should be replaced with real schedules from ASI and Boring as soon as available; the fintech number in particular is bimodal and warrants two instrument classes. The per-payload cost falls linearly with packing density, so the entire Web 2.0 case strengthens with the elastic-hashing gains the rent note quantifies—for low-fee consumer volume, dense packing is the difference between profit and loss, not a refinement. The compliance multipliers are coarse stand-ins for replication, audit, and dedicated-infrastructure costs and should be replaced with per-sector infrastructure models before any commitment. Finally, the single largest non-modelled risk is customer concentration—ten artists in music, one federal buyer in medical—which the portfolio sequencing above is in part designed to diversify.

## References

- [1] F1R3FLY Industries. Rent and shard splitting: a storage-economics companion to occupancy-driven sharding. Working note, 2026.
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